Can the search tilt? Geometry as an explanation for Bayesian optimisation versus CMA-ES

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Repository: [TODO: https://github.com/USER/REPO]

Abstract

We test one explanation for a difference between Bayesian optimisation (BO) and CMA-ES: on a rotated, ill-conditioned problem, what matters is whether a method's search geometry can tilt away from the coordinate axes, not which family it belongs to. We run a 2×2 ablation: full against diagonal covariance for CMA-ES, and a full kernel metric against the standard ARD kernel for BO. Diagonal CMA-ES fails only when the valley lies between the axes. ARD cannot even represent the tilt. A first head-to-head showed *no* rotation penalty for BO. We traced that negative result to a fitting failure that made BO stall on every problem. Once it was fixed, BO-ARD paid a rotation penalty of +2.0 decades at $\kappa = 10^4$ while the full metric did not, the same pattern as diagonal against full CMA-ES. We then test which of CMA-ES's theoretical invariances survive in practice (rank invariance does, bit for bit; units invariance only with a matched initial covariance) and where the box boundary, a curved valley and noise change the comparison. Under noise, the rule for choosing the final point mattered as much as how the budget was split. With each family's best choices the two are level. Nothing here is specific to one family: the axis restriction, not the algorithm, predicts the rotation failures.

1 Question and explanation under test

CMA-ES adapts a full covariance matrix from the ranking of sampled candidates, so its sampling distribution can stretch and *rotate* to match the local Hessian [5]. BO fits a Gaussian-process surrogate and chooses points through an acquisition function [4]. Its standard squared-exponential kernel with automatic relevance determination (ARD) uses a squared distance $\sum_i (x_i - x'_i)^2/\ell_i^2$, a *diagonal* metric. Trust-region BO inherits the same restriction, because TuRBO's trust region is a hyperrectangle whose side lengths are set per coordinate from the ARD lengthscales [3].

Explanation under test. On a rotated, ill-conditioned problem, performance depends on whether the method's search geometry *can tilt*. That property, not the family, decides the outcome:

- *can tilt:* full-covariance CMA-ES, and a GP with a full metric $M = LL^\top$ in $k(x, x') = \exp(-\frac{1}{2}(x - x')^\top M(x - x'))$;
- *cannot tilt:* diagonal CMA-ES, and the ARD kernel (M diagonal).

Conditioning, rotation and curvature. Near a minimum, $f(x^* + h) \approx f^* + \frac{1}{2}h^\top Hh$. The eigenvalues of the Hessian H are the *curvatures* of f along its eigenvectors. The *condition number* $\kappa = \lambda_{\max}/\lambda_{\min}$ is the ratio of the largest curvature to the smallest, so an ill-conditioned problem is a narrow valley: steep across, flat along. The eigenvectors give the valley's *orientation*; a rotation changes them but leaves κ unchanged. This is why our problems vary κ and rotation separately. On a quadratic, H is the same everywhere, so one tilt describes the whole problem. A well-adapted CMA-ES covariance approximates H^{-1} up to scale [5], and a full GP metric can align with the eigenvectors of H. A step that ignores the orientation is a poor one on a narrow, tilted valley. Along the valley it must stay short enough not to climb the steep walls, so progress along the flat direction is slow. Rosenbrock (Section 7) adds a third property: its Hessian changes from point to point. The valley floor *bends*, so the local orientation and local κ change along the path. This separates *local* geometry, which CMA-ES re-adapts as it moves, from *global* geometry, the single metric a stationary kernel learns for the whole box.

The explanation makes three falsifiable predictions. (i) Each axis-restricted method pays a penalty under rotation that its tilting counterpart does not. (ii) The penalty appears only when the problem is both ill-conditioned and rotated, because an aligned problem needs no tilt. (iii) The tilting variant pays a modest price when no tilt is needed, because it has more parameters to learn. A brief from the course also asks which theoretical invariances survive the implementation, and how changes of units and box constraints behave; Sections 5–7 take these up.

2 Methods and experimental design

Problems. Ellipsoids $f(x) = \frac{1}{2}(x - x^*)^\top QDQ^\top(x - x^*)$ with log-spaced eigenvalues from 1 to κ, following COCO practice [7]: the optimum x^* is drawn on a radius-2 sphere (never at the start point), Q is the identity (aligned), a fixed angle, or a Haar-random rotation, and each *instance* fixes Q and x^* from its index. Section 7 adds Rosenbrock (COCO f8/f9 [6]) and additive Gaussian noise.

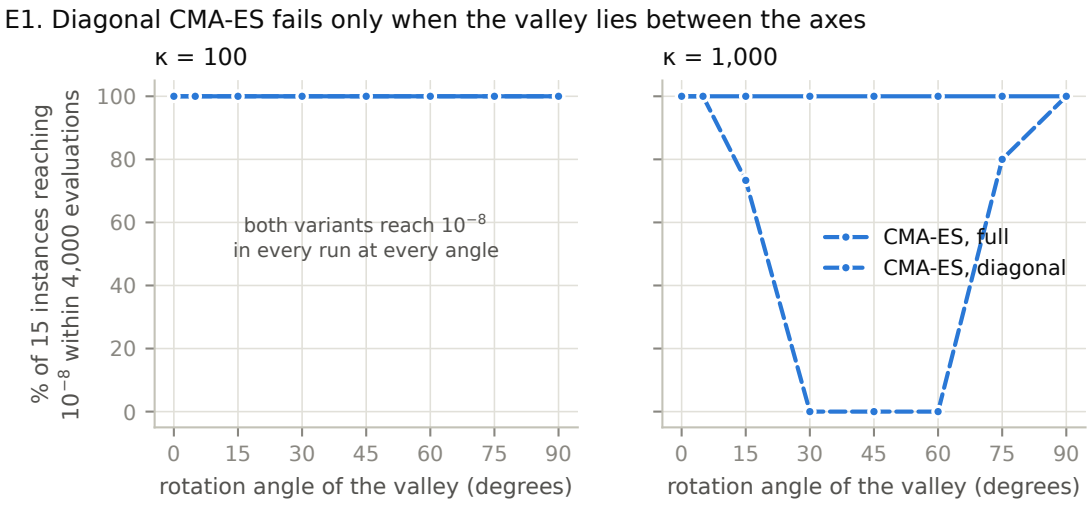


Figure 1. E1. Success of full (solid) and diagonal (dashed) CMA-ES against the angle of a 2-D valley.

Methods. CMA-ES follows Hansen's tutorial (rank-one and rank-μ updates, cumulative step-size adaptation, default population; $\lambda = 6$ in 2-D). The diagonal variant zeroes the off-diagonal covariance entries after every update. BO is GP-EI: an initial uniform design of $2d + 4$ points, EI maximised over 200 uniform candidates and then polished with L-BFGS-B inside the box. Targets are log-transformed and standardised before fitting. GP hyperparameters are fitted by MAP with BoTorch's dimension-scaled LogNormal lengthscale prior [8], with 10 restarts and exact marginal-likelihood gradients [9, eq. 5.9]. We place the prior on the *eigen*-lengthscales of M, so that it does not favour axis-aligned metrics, which would build in the restriction under test. All four methods are implemented in NumPy/SciPy in the repository, so every kernel and update choice is visible.

Budget rule (shared by E2–E5). Cost is counted in objective evaluations, by wrapping the objective; the initial design counts. Both families get the same box $[-5, 5]^d$: BO searches it, and CMA-ES starts at its centre with $\sigma_0 = 0.3\times$ box width and is not otherwise constrained (the asymmetry is examined in Section 6). The cap is 200 evaluations. Every evaluation is logged, so each run gives both a fixed-budget view (f_{best} at 20/50/100/200) and runtimes to targets $10^{-1}, 10^{-3}, 10^{-5}$, with failures reported as success rates rather than dropped [7]. Wall-clock time is logged because it separates the families sharply (about 0.002 s per CMA-ES run against 120–260 s per BO run). Instance index = seed for every method, so cross-method comparisons are paired; we report median differences in $\log_{10} f$ ("decades") with 95% bootstrap intervals over 15 instances. Aligned-against-rotated comparisons are unpaired, because the rotation is drawn together with the optimum.

Reproducibility. Each experiment is one script in `code/` that writes CSVs to `results/`. Section 8 of the notebook reloads every number quoted here, and `code/make_figures.py` rebuilds every figure from the CSVs. BO runs are bit-reproducible when BLAS is single-threaded, which the scripts enforce. With multithreaded BLAS, two runs of one seed differed by 6×10^{-5} in relative terms, and the difference compounds; results are reproducible in distribution across machines, not run for run (details in `readme.md`).

3 The tilt in each family

E1: CMA-ES. At $\kappa = 1{,}000$ in 2-D (15 instances, 4,000 evaluations, target 10^{-8}), diagonal CMA-ES solves every instance at 0° and 90° and none between 30° and 60°; full CMA-ES solves every angle (Fig. 1). The 90° point is the control: it only swaps the axes, so recovery there shows that the rotation, not the conditioning, causes the failure. At $\kappa = 100$ both variants succeed at every angle, as prediction (ii) requires. Under Haar-random rotation full CMA-ES needs 405 against 408 median evaluations (30 instances), the rotation invariance claimed by Hansen [5], while diagonal success drops to 47%, and to 0% by $d = 16$. On aligned problems the full variant costs up to 21.6% more evaluations at $d = 32$, because it learns $d(d + 1)/2$ covariance entries instead of d: prediction (iii).

E2c: the surrogate. Rather than infer the kernel's geometry from optimisation performance, we read it directly. We fit both kernels to the same 40-point uniform designs (10 instances) and measure the angle between the learned long axis and the true flat direction (Fig. 2). ARD is off by exactly 45° on 45° problems at every κ, and by a median 18.1° under random rotation; the full metric is within 0.2°–4.1°. On rotated problems ARD's held-out RMSE is 1.3–1.9× the full metric's from $\kappa = 100$. On aligned ones ARD is better (ratio 0.75–0.93 at $\kappa = 10$–1,000), the surrogate-side counterpart of the CMA-ES overhead.

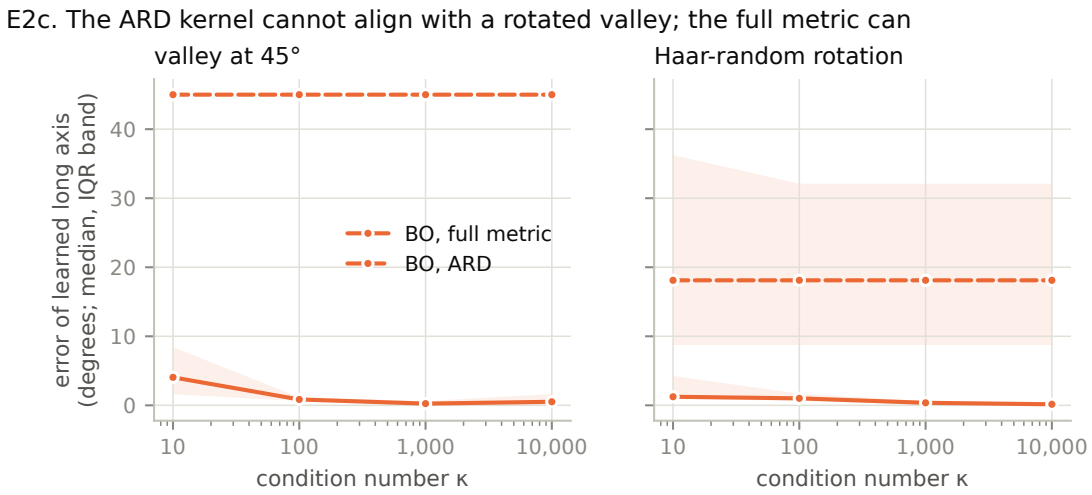


Figure 2. E2c. Error of the GP's learned long axis (median over 10 instances; band = interquartile range).

An earlier maximum-likelihood fit had a hidden bound $\ell \leq e^4$: the long lengthscale sat on it in 60–80% of fits at $\kappa \geq 10^3$ (E2d), so any lengthscale *ratio* we reported was set by the bound, not by the data. The MAP prior removed every bound hit and left the axis errors unchanged. We therefore report direction and RMSE, not ratios.

4 Head to head, a negative result, and its cause

E2, first run. Under the budget rule (2-D; $\kappa \in \{1, 100, 10^4\}$; aligned and random rotation; 15 instances), BO led CMA-ES by 0.6–1.0 dec at 20 evaluations, and CMA-ES led by 2.3–3.8 dec at 200 for $\kappa \leq 100$. On rotation, however, prediction (i) *failed* for BO: BO-ARD's rotation penalty at $\kappa = 100$ was +0.05 [-0.53, +0.63] decades, while diagonal CMA-ES paid +1.70 [+0.41, +2.57]. A second anomaly stood out: BO stalled near $10^{-2.6}$ even on the round bowl ($\kappa = 1$), the easiest case for a GP, and no BO run reached 10^{-5}.

E2e, diagnosing the stall. We re-ran BO on the round bowl with per-iteration logging, behaviour unchanged, and tested four hypotheses, each with its own signature:

- *Floor in the log transform:* rejected. The best value stayed 54–82× above it.
- *The acquisition search misses the improving region:* rejected. A dense search around the incumbent found no more EI than the random candidates did.
- *EI underflow:* the mechanism. After 50 evaluations the largest EI anywhere was below 10^{-12} in 97–100% of iterations. The chosen points landed a median 3.8 from the incumbent in a box of width 10, so BO was effectively sampling at random.
- *The GP explains the optimum away as noise:* the cause. The fitted noise SD was 0.2–0.4 (standardised units) on a noise-free objective. The posterior mean at the incumbent sat about 1.5 units above its observed value, 2–2.5× the gap to the second-best point. EI, measured against an incumbent the model considers unreachable, collapses.

Changing one thing at a time (BO-ARD, same instances, paired), fixing the noise variance at 10^{-6} improved the median by −1.43 [-1.83, -0.73] decades at 50 evaluations and −1.17 [-1.71, -0.74] at 200; every run then reached 10^{-3} (27% before). LogEI [1], which fixes the underflow but not its cause, changed nothing (+0.01 [-0.79, +0.47]). Lowering the transform floor as well added a non-significant −0.28 [-0.68, +0.15], so we adopted the fixed noise alone. Even fixed, BO stops improving near 10^{-4}. The fixed-noise logs explain why: the acquisition search now fails, the hypothesis rejected before the fix. After 100 evaluations the GP's shortest lengthscale is about 0.07, so the region worth searching is about 0.1 wide. In 98–99% of iterations a dense search there finds $> 10\times$ (median about 800×) the EI of the point chosen, while only 1–2% of chosen points land within 0.1 of the incumbent: 200 uniform candidates in a 10×10 box almost never hit it. Adding candidates around the incumbent, as BoTorch does, is the natural remedy.

E2 with the fix. We re-ran the 180 BO runs both ways on one machine, fitted noise and fixed noise (CMA-ES is unaffected), so the comparison is paired. The fix lifted BO significantly in 10 of 12 condition, geometry and kernel cells, by 0.68–1.50 decades at 200 evaluations. **The negative result reversed** (Fig. 3). With fitted noise, neither kernel showed a rotation penalty on this machine either (BO-ARD at $\kappa = 10^4$: +0.04 [-0.27, +0.71]). BO-ARD's rotation penalty became +0.54 [+0.24, +1.41] decades at $\kappa = 100$ and +2.01 [+1.28, +2.97] at $\kappa = 10^4$;

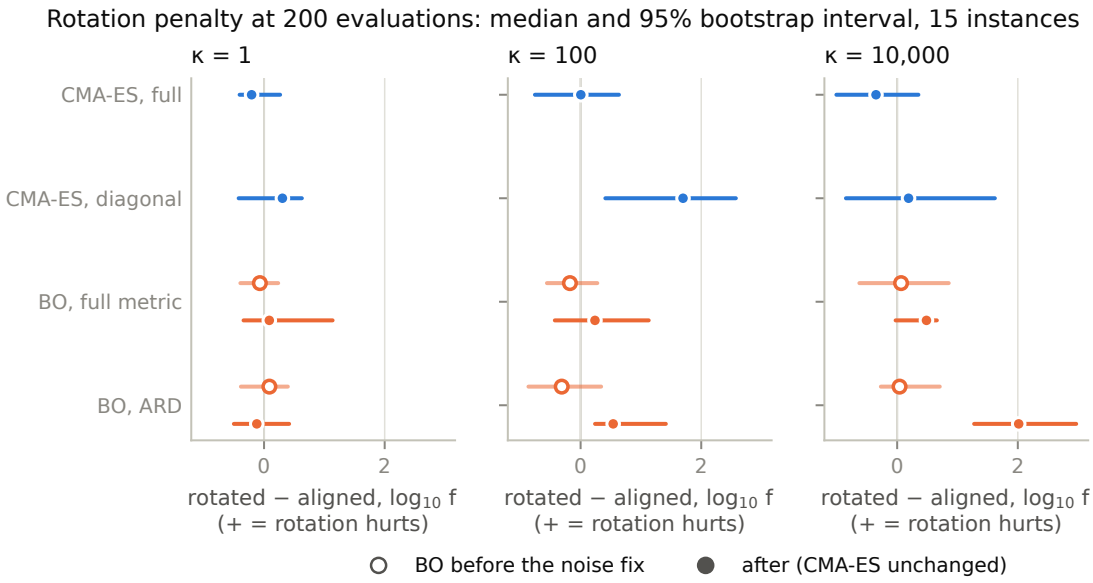


Figure 3. Rotation penalty at 200 evaluations (median and 95% bootstrap interval). Hollow: BO before the noise fix (same machine, paired instances). The axis-restricted variant pays in both families once BO is no longer stalled.

BO-full's stayed non-significant (+0.24 [-0.43, +1.13], +0.49 [-0.03, +0.66]). Paired on the same rotated $\kappa = 10^4$ instances, BO-ARD ended 1.40 [+0.68, +1.69] decades behind BO-full; on aligned ones the two were level (−0.13 [-1.15, +0.19]). This is the CMA-ES pattern in a different family: the stall had been hiding the ARD penalty, because no method can lose ground it never gains.

Across families (Fig. 4), BO-full now leads CMA-ES-full by 0.85–1.25 decades at 50 evaluations in every cell, and on aligned $\kappa = 10^4$ at every budget (−1.32 [-1.92, -0.62] at 200). CMA-ES still wins by 200 evaluations at $\kappa \leq 100$ (by 1.2–2.7 decades), where its linear convergence overtakes BO's stalled progress. BO's early lead costs about 10^5 times more computation per run.

5 Which invariances survive the implementation?

E3 (Fig. 5; aligned $\kappa = 100$; paired with an untransformed baseline; every evaluation scored on the original f). Hansen lists invariance to strictly increasing transformations of f, and to invertible linear maps of the search space *provided the initial mean and covariance are mapped with it* [5].

- *Monotone maps of f* (log1p, cube, square root, $f + 1$). Both CMA-ES variants produced runs bit-identical to the baseline in all 120 cases. BO is not invariant, and the direction depends on the map. Cubing costs +1.72 [+1.14, +2.22] (ARD) and +1.74 decades (full), and no run reaches 10^{-3}; log1p helps BO-full (−0.52 [-0.98, -0.03]). The cube result has a precise cause in our implementation: the target floor is $10^{-6}\times$ the spread of observed values, and cubing stretches the spread so much that the floor corresponds to $f \approx 14$, below which BO sees nothing. The floor is not the whole story, though: square root makes it far finer (10^{-9}) without helping.
- *Units* ($u_i = x_i/s_i$). With default set-ups, no method is invariant: $s = (1, 10)$ raises the effective conditioning to 10^4 and costs CMA-ES about +3.9 decades and BO +1.3–1.4. Mapping CMA-ES's initial covariance as the theory requires ($C_0 = S^{-2}$) restores invariance: exactly (to 10^{-12}) for the diagonal variant, and in distribution for the full one (+0.21 [-1.07, +0.85]). The full variant is not identical run for run, because its eigendecomposition sorts the axes by eigenvalue and so reassigns the random draws. BO's equivalent, normalising each input to its own range [2], is invariant by construction, but our box has one width for all coordinates, so it was not run.

6 The box boundary

E4 (Fig. 6; 2-D aligned, $\kappa \in \{1, 100\}$). Because every method starts at the centre, an optimum near an edge is also far from the start. We therefore compare each edge position with an interior control at the same distance ($(\pm 3.5, \pm 3.5)$, 1.5 from both faces).

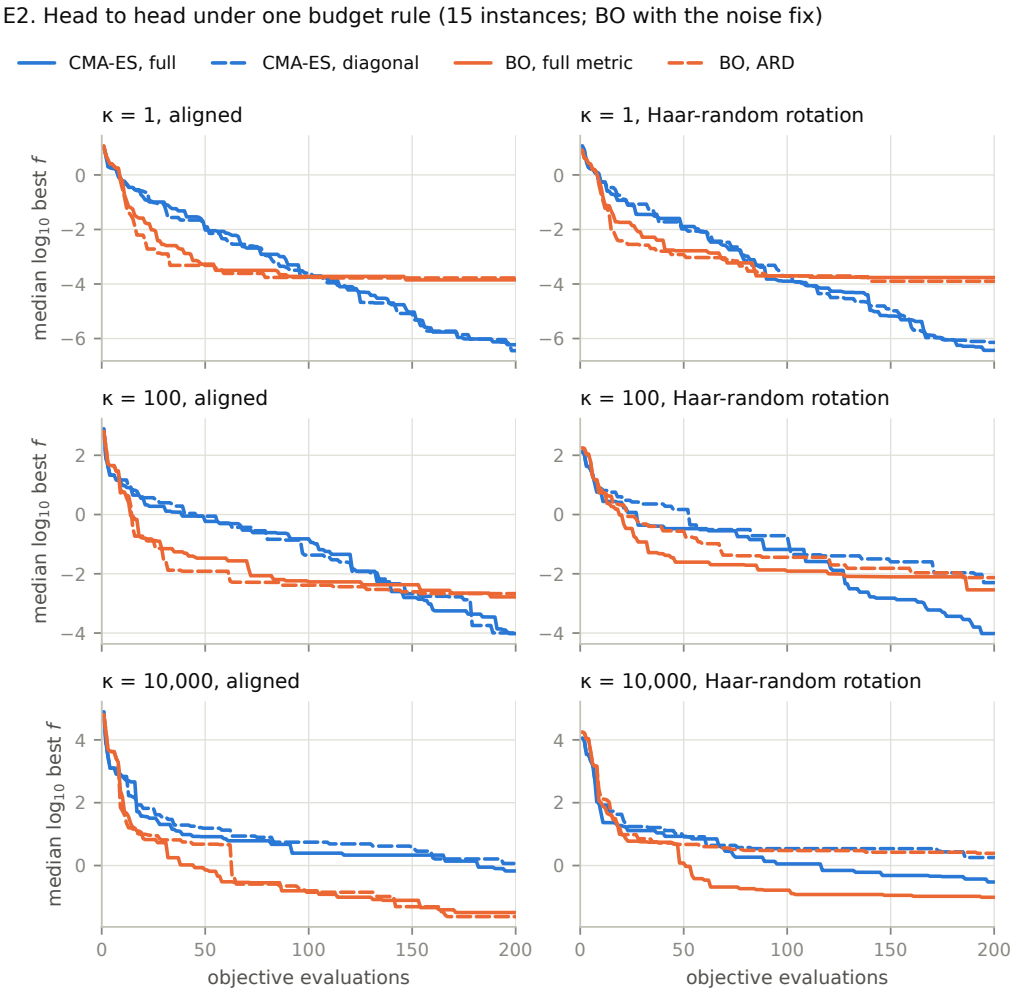


Figure 4. E2 with the BO fix: median best-so-far over 15 instances. Colour = family; dashed = axis-restricted.

- *BO* is worse when the optimum lies 0.1 inside the edge (+1.62 [+0.57, +2.10] at $\kappa = 1$, +0.73 [+0.31, +0.99] at $\kappa = 100$) and better when it lies exactly on it (−0.95, −1.68, both significant). BO puts 9–15% of its evaluations exactly on the boundary, consistent with GP-EI's known over-exploration there [10]: the posterior variance is largest where there is no data beyond the edge, and the bounded EI optimiser stops exactly on it. With the optimum in a corner, BO evaluates it exactly in 93–100% of runs, at a median of evaluation 9–10. That is an artefact of this preference, not evidence of better search.
- *Unconstrained CMA-ES* stays within 0.53 decades of the control everywhere; it is affected by distance from the start, not by the edge. But 47–63% of its evaluations fall outside the box when the optimum is on an edge or in a corner. A reflection boundary transform (a simplified version of pycma's default) removes the violations at no significant cost, except when its periodicity lets the mean drift onto a mirror copy of an interior optimum (+1.04 [+0.43, +1.64]).

7 Beyond the quadratic

E5 (Fig. 7). We wrote two predictions down before running.

- *Rosenbrock (f8/f9).* *Why this function:* every other problem here is a quadratic, whose Hessian, and so whose orientation and conditioning, is the same everywhere. Rosenbrock is the standard benchmark (COCO f8/f9) that is still unimodal and smooth, but whose valley floor *bends* along the parabola $z_2 = z_1^2$ (Section 1). It changes one thing, the constancy of the geometry, and so tests whether the explanation needs a *local* tilt or only a global one. *Why it needs a longer budget:* a method must travel along the bending floor, and its learned orientation goes stale as the floor turns. A step that is long along today's direction climbs the wall further on, so steps stay short and the shape must be re-learned repeatedly. Near the optimum the valley is also narrow (local $\kappa \approx 2{,}500$). Full CMA-ES needs a median of about 500 evaluations to reach 10^{-5} (E9 below), so 200 evaluations end before the final stage. *At 200 evaluations*

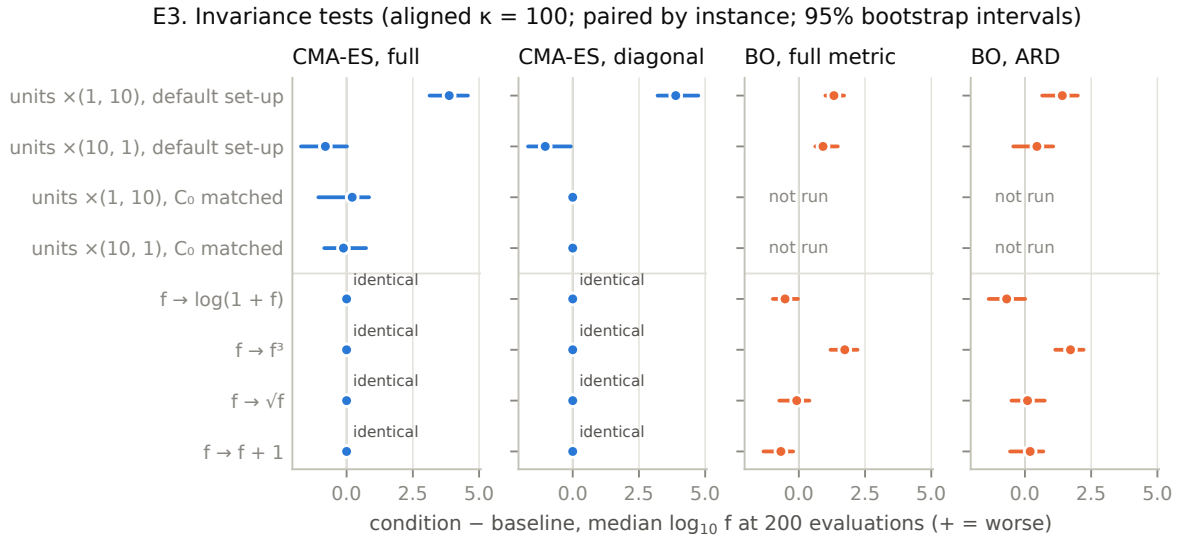


Figure 5. E3. Condition minus baseline, median $\log_{10} f$ at 200 evaluations (+ = worse); "identical" = all 15 runs bit-identical to the baseline.

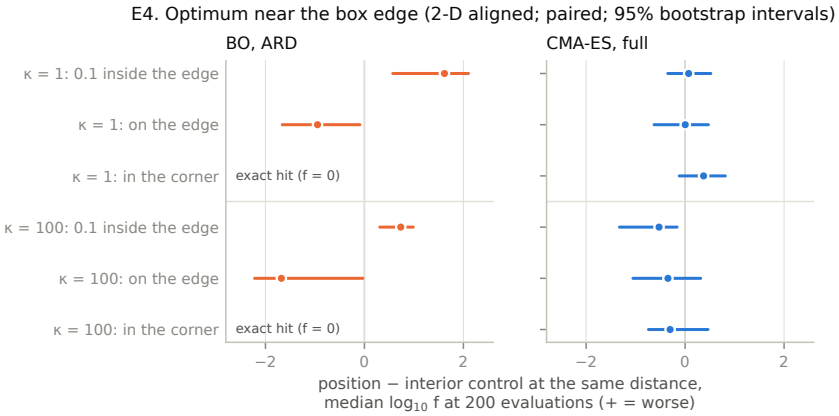


Figure 6. E4. Edge position minus a same-distance interior control (+ = worse).

(E5) no method converges ($10^{-1.35}$–$10^{-0.49}$). Even "aligned" Rosenbrock has its valley at 63.5° to the axes near the optimum, so rotating it changes little. BO-full beats BO-ARD by 0.63 [0.21, 1.08] (f8) and 0.39 [-0.24, 0.96] (f9) decades, against 1.40 on the rotated quadratic, which is weak support for the "one global tilt" prediction only. *With 4,000 evaluations (E9, CMA-ES only)* the picture is sharp. Full CMA-ES reaches 10^{-5} in every run (median 535 and 492 evaluations), while diagonal CMA-ES reaches it in 0% of f8 runs and 27% of f9 runs, about 3,500 evaluations behind (paired, failures counted at the cap). We had predicted that both would converge. Diagonal fails even on the "aligned" f8, because what matters is the valley's *local* orientation, which is tilted near the optimum whatever the label says.

- *Noise* (additive, $\sigma \in \{0.01, 1\}$; each method recommends its lowest observed point, scored on the noise-free f, as the course brief requires). The prediction was that the noise fix would backfire once the objective is noisy. It did not: fixed-noise BO still beat fitted-noise BO (-0.41 [-0.73, -0.10] at 200, $\sigma = 0.01$). Noise removes CMA-ES's late lead (from $+1.35$ noise-free to $+0.42$ and -0.01). It misleads CMA-ES's recommendation more than its search: at $\sigma = 0.01$ it evaluates points at $10^{-3.57}$ but recommends one at $10^{-2.40}$.

8 Noisy evaluations: budget split and recommendation

E6 (Fig. 8) follows E5's noise test with the two questions the course brief poses. How should a noisy budget be divided between new points and repeats, and which point should be recommended? Setting: the aligned $\kappa = 100$ quadratic, additive noise $\sigma \in \{0.01, 0.1, 1\}$, 200 noisy evaluations per run with repeats counted, and full CMA-ES and BO-ARD with the fix. Each recommendation is scored on the noise-free f. The rules are R1, the lowest observed value; R2, the model's estimate (CMA-ES's final mean, or the point with the lowest posterior mean of

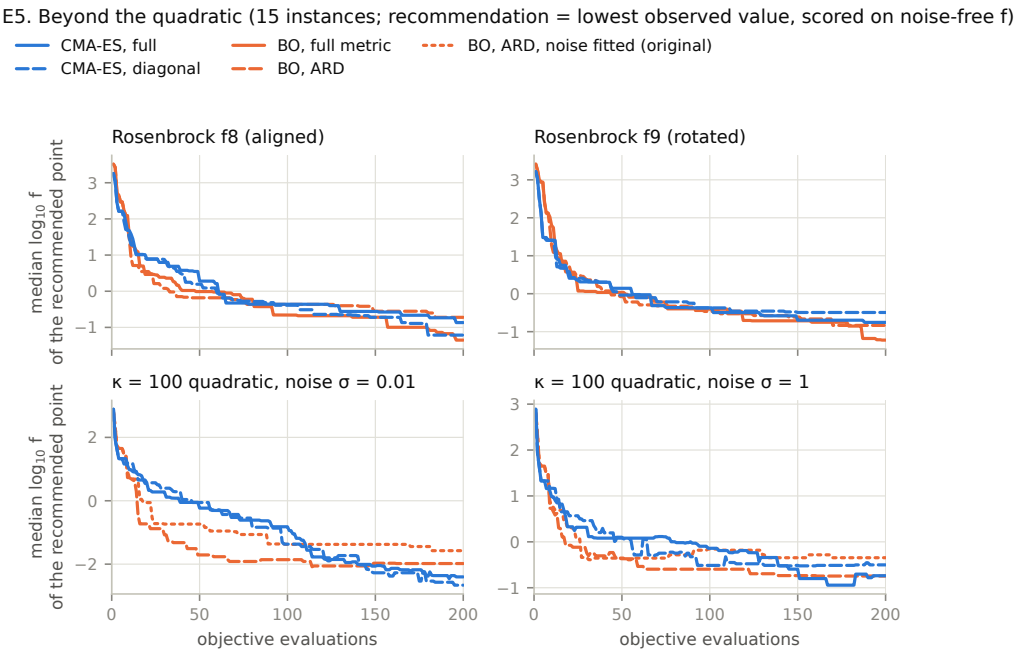


Figure 7. E5. Median noise-free value of the recommended point. Dotted: BO-ARD with the noise fitted (original).

a GP refitted *with noise estimated*); and R3, the best average after re-evaluating the top five points.

- *Rules.* For BO, R2 beats R1 by 0.30 [0.15, 0.78] decades at $\sigma = 0.1$ and by 1.08 [0.28, 1.29] at $\sigma = 1$: the search was fine, but picking the lowest reading threw it away. For CMA-ES, the mean helps only with a larger population ($\lambda = 24$: 0.47 and 0.33 decades, both significant). With $\lambda = 6$ it ties with R1, because a mean of three selected points averages little.
- *Splits.* Repeats did not pay at this budget. Resampling each CMA-ES candidate 4× leaves 48 distinct points and costs 1.1–3.0 decades. BO replicates cost 1.6 and 1.1 decades at $\sigma \leq 0.1$ and do not help at $\sigma = 1$ (+0.84 [-0.14, 1.68]), though they cut BO's computation 27-fold (5 s against 141 s). A final reserve of 40 re-evaluations never helped significantly. A 4× larger population, which averages without repeating, was the only split that helped (CMA-ES at $\sigma = 1$: −0.57 [-0.85, -0.08]), and it costs +0.62 at $\sigma = 0.01$.
- *Across families.* With each family's best configuration, BO and CMA-ES are level at every noise level (+0.02, +0.08, −0.18 decades; every interval contains 0). Three of our five written predictions failed or held only in part (`code/run_e6_noise_budget.py`).

At this budget the best use of noisy evaluations was to keep exploring new points and let a model average at the end. One anomaly is unexplained: with replicates, BO's R2 is *worse* than R1. A plausible cause is that the averaged data are nearly noise-free, so the noise-fitting GP fails as in Section 4.

9 What is established, and what is not

Established. (1) On rotated ill-conditioned quadratics, the axis restriction predicts failure in both families: diagonal CMA-ES and ARD-BO pay, while their tilting counterparts do not, and the penalty vanishes on aligned or well-conditioned problems. (2) Tilting has a modest cost when no tilt is needed (up to +21.6% evaluations for CMA-ES; worse held-out RMSE for the full GP metric on aligned problems). (3) CMA-ES's rank invariance survives the implementation exactly; its linear invariance holds only under the stated condition on C_0. BO has neither invariance, and how f is written can move it by 1.7 decades. (4) The box is part of BO's model and invisible to CMA-ES; a fair comparison has to state the bound handling for both. (5) Under noise at this budget, model-based recommendation (GP posterior mean; a large-population CMA-ES mean) beats the lowest reading, and repeating evaluations does not pay.

Uncertain or limited. BO was run only in 2-D, with one kernel family, so the dimension trend of the ARD penalty is inferred from CMA-ES (E1), not measured. BO stops near 10^{-4} even after the fix; the logs point to the acquisition search (Section 4), and the comparison at 200 evaluations depends on it. The full GP metric is overconfident (worse log predictive density than ARD nearly everywhere), and about 2% of fits still miss the best hyperparameter optimum. CMA-ES ran without restarts. A check (E7) shows this does not matter here: full CMA-ES solves every quadratic without them, and IPOP-style restarts (4 runs, population doubling from

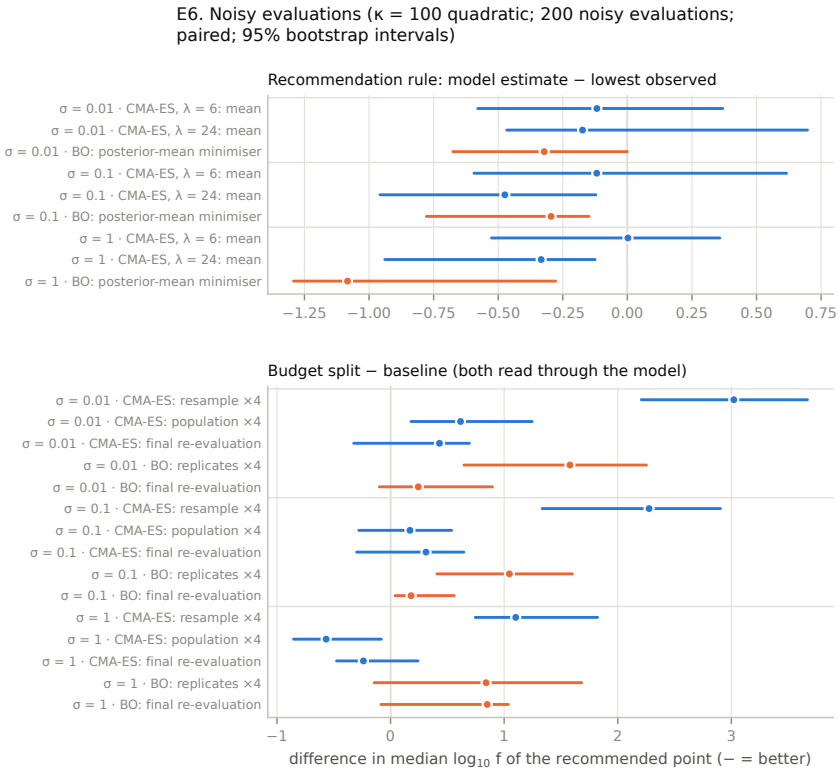


Figure 8. E6. Top: model-based recommendation minus lowest reading on the same runs. Bottom: each budget split minus the baseline. Negative = better.

6 to 48) leave diagonal CMA-ES's rotated $\kappa = 10^4$ failure in place (7% reach 10^{-5} against 20% for one long run), because a restarted diagonal run still cannot tilt. Across machines, BO results agree in distribution but not run for run: re-running the original E2 elsewhere matched 40–93% of BO-ARD runs and 0–27% of BO-full runs to 0.01 decades, and one cell's median moved by 0.6 decades. All before/after comparisons in the report are therefore paired on one machine. With noise, the geometry argument predicts little at this budget. Several of our written predictions failed or held only in part (E5, E6, E9) and are reported as such.

For algorithm choice. The question "BO or CMA-ES?" is partly the wrong one. Whether the method can represent the problem's orientation matters as much as the family. A practitioner facing a rotated, ill-conditioned problem should avoid an ARD kernel or a diagonal ES for the same reason. Under noise, the choice of recommendation rule matters as much as the choice of family.

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